MedBhasha Round 2 — Technical Report

Team JUSL Health Home · COMSYS Hackathon-8 · Annotate → NER → Cluster → Retrieve

1. Summary

We built an end-to-end, **offline and deterministic** Bengali clinical NLP pipeline and packaged it as one runnable notebook (`MedBhasha_Pipeline.ipynb`):

1. **Data.** After the organisers' feedback (40 notes / 612 tokens in Round 1, target ≈ 10 000 tokens) we grew our own annotated corpus to **232 notes / 12060 tokens / 2899 entity spans** covering 23 clinical domains and all 10 entity types (rarest type: PROCEDURE = 73 spans). The corpus passes the official `validate.py` with 0 errors and 0 warnings.
2. **NER.** A lexicon-augmented, section-aware linear-chain CRF. Cross-validated entity-level F1 on our own data: **0.876** (random 5-fold) and **0.861** (whole clinical domains held out).
3. **Clustering.** Unsupervised clustering of the 1691 released notes from their predicted entities: **90 clusters** (ward, chosen by internal metrics only), silhouette **0.777**, Davies–Bouldin **0.717**.
4. **RAG.** Retrieval over our annotated notes (+ the released notes annotated by our own NER model) with grounded, extractive Bengali answers and `[note-id]` citations for the 30 test queries; a live-query function for the finale. No external API, no web retrieval, no pretrained generator.

Three caveats we want the reader to see first.

- All our training data are *synthetic and written by a single author*; no clinician reviewed the annotations. Cross-validated scores are therefore an **upper bound** — the hidden test F1 will very likely be lower, especially if the organisers' gold uses other span conventions (§4.6).
- The RAG rule says "retrieve from your own annotated corpus only". The queries ask for *patients*, which only the released notes can supply. We therefore provide **two answer files** — released notes annotated by our own NER (default) and strictly team-corpus-only — and ask the organisers to confirm the intended reading (§6.5).
- We looked at a small number of released notes and the header statistics to learn the *format* of the notes (sections, shorthand headers such as `C/O`, `Dx`, `Rx`). We did not copy, annotate or train on any released text (verified: maximum character-5-gram Jaccard between any training note and any released note is **0.28**).

2. Rule compliance at a glance

Rule / requirement	How we comply
Approved sources only; source declaration	100 % synthetic team-authored notes;

Rule / requirement	How we comply
with annotation method	`round1_deliverables/SOURCE_DECLARATION.md` lists every batch and states the annotation method (no rule-based tool, no clinician review).
Entity schema + BIO format, mandatory `validate.py` report	`data/round1/annotated_corpus.csv` (columns `note_id,token,tag`); report in `round1_deliverables/validation_report.txt`; prediction file validated too (`outputs/ner_predictions_validation_report.txt`).
EDA summary	`round1_deliverables/EDA_Summary.pdf` (+ §3.4 below).
NER trained on the team's **own** Round-1 annotations	CRF trained only on `annotated_corpus.csv`; **no pretrained model, no external data, no use of the released notes for training**.
Clustering unsupervised, features from NER output	Representation uses predicted entities only; model selection by silhouette / Davies–Bouldin only.
RAG: team's own corpus only, no external API / web	Everything local; knowledge base = our notes (+ released notes annotated by our NER); strict variant provided.
One runnable notebook; exact reproduction	`MedBhasha_Pipeline.ipynb`; fixed seed 42; every output hashed in `outputs/run_manifest.json`; `scripts/check_reproducibility.py` re-runs everything under a different `PYTHONHASHSEED` and compares hashes.
Cite tools / pretrained models	No pretrained models. Libraries: NumPy, pandas, SciPy, scikit-learn, CRFsuite via `sklearn-crfsuite`, `seqeval`, NetworkX, Matplotlib. Annotation method: see the source declaration.

3. Data

3.1 From Round 1 to Round 2

The Round-1 submission had 40 notes (612 tokens) and most entity types had < 30 mentions. The organisers recommended ≈ 10 000 tokens, prioritising rare entity types, and noted that few-shot LLM approaches work better on tiny corpora. We did the following:

- kept the 40 Round-1 notes (`n001`–`n040`) and **fixed 8 inconsistent tags** (`round1_fixes_log.tsv`), e.g. dosage forms such as ইনহেলার moved into the MEDICATION span, ফলো-আপ removed from TEMPORAL;
- wrote **192 new notes** (`n041`–`n232`) in 9 batches, deliberately mixing registers: long structured outpatient notes, admission/discharge summaries, shorthand notes (`C/O … O/E … Dx … Rx … Adv … F/U`), and very short informal first-person notes;
- repaired the metadata file: the Round-1 `note_metadata` covered only 20 notes with stale domain labels; the new `data/round1/note_metadata.csv` covers all 232 notes.

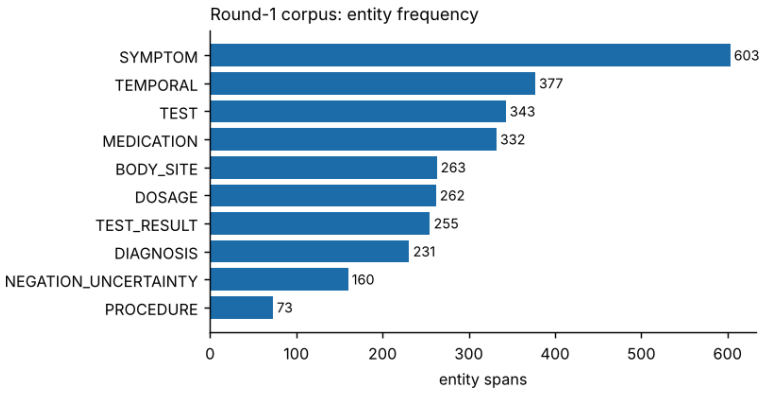
3.2 Annotation process

Notes and entity spans were written as inline markup (`{DX|ডায়াবেটিস}`), a deterministic script aligned spans to tokens and produced BIO tags (it refuses spans that cut tokens, overlap, or contain punctuation). No clinician reviewed anything; there is **no inter-annotator agreement**. All decisions are documented in `docs/ANNOTATION_CONVENTIONS.md`. Source category: *synthetic clinical notes generated by the team* — no real patient data, no EHR, no web text.

3.3 Corpus statistics

entity type	spans	distinct forms	notes containing	share %
SYMPTOM	603	221	206	20.8
TEMPORAL	377	114	193	13.0
TEST	343	144	162	11.8
MEDICATION	332	184	173	11.5
BODY_SITE	263	147	169	9.1
DOSAGE	262	135	153	9.0
TEST_RESULT	255	188	151	8.8
DIAGNOSIS	231	179	189	8.0
NEGATION_UNCERTAINTY	160	20	123	5.5
PROCEDURE	73	50	59	2.5

Notes: 232 · tokens: 12060 (mean 52 per note, median 62, range 9–103) · spans: 2899 · domains: 23 (largest: Respiratory 30, Gastroenterology 27, Cardiology 26, Dermatology 23).



Entity frequency in our corpus

3.4 EDA findings (what the numbers told us)

- `SYMPTOM`, `TEMPORAL`, `TEST` and `MEDICATION` dominate; `PROCEDURE` is rarest (73 spans) because most clinical acts in outpatient notes are tests or drugs. This is the schema's nature, not a labeling gap.
- Note length is bimodal by design (very short informal notes vs. ≈ 100-token structured notes); the released notes are all structured and ≈ 131 tokens long, so the structured part of our corpus matters most.
- Co-occurrence (figure `outputs/figures/eda_cooccurrence.png`): P(DOSAGE present | MEDICATION present) = 0.88, P(TEST_RESULT | TEST) = 0.92, P(TEMPORAL | SYMPTOM) = 0.90, P(NEGATION_UNCERTAINTY | DIAGNOSIS) = 0.56 — the footprint of prescription and examination templates (descriptive, not clinical findings). The EDA doubled as a self-check: it exposed a split-vs-merged inconsistency for "organ + symptom" phrases (পেটে ব্যথা as one SYMPTOM vs. `BODY_SITE+SYMPTOM`), which we resolved in favour of the organisers' worked example (BODY_SITE F1 rose by about 9 points under domain hold-out in our development runs after the fix).

3.5 Diversity and originality checks

- exact duplicates: none; maximum character-4-gram Jaccard between any two of our notes: **0.45** (near-duplicate threshold 0.6);
- similarity to the released test notes: max char-5-gram Jaccard **0.28** (unit test `tests/test_basics.py::test_no_note_copied_from_release`);
- the style of batches 03–11 imitates the *genre* of the release (section headers, shorthand headers) — we disclose this openly; no sentence was copied.

4. Stage 1 — Named-entity recognition

4.1 Model

A linear-chain **Conditional Random Field** (CRFsuite via `sklearn-crfsuite`; L-BFGS, c1 = c2 = 0.05, 200 iterations, all transitions allowed). Why a CRF: with ≈ 12 k annotated tokens a

neural encoder over-fits, and pretrained Bengali encoders (IndicBERT / MuRIL / BanglaBERT) cannot be downloaded in an offline, exactly-reproducible run. A CRF trains in seconds and is bit-for-bit deterministic. We did **not** use an LLM few-shot approach because it requires a hosted or very large local model, which violates the no-external-API / exact-reproduction constraints.

Features per token: word, 1–4-char prefixes and suffixes (Bengali inflection and drug morphology), orthographic shape (digit / `/` / `.` / `%` / Latin), ±2-token context window and bigrams, **gazetteer marks** and **note-section context**.

- **Gazetteer** (`src/medbhasha/lexicon.py`, 762 terms): bilingual drugs, diseases, symptoms, tests, body sites, procedures, negation and dosage words, written from general clinical knowledge — *not* mined from the released notes. Matching is spelling-normalised (hasanta removal, ড়/য় composition, case-ending stripping).
- **Section context** (`ner.py::section_context`): the nearest preceding header (নির্ণয়`:`, ব্যবস্থাপত্র`:`, ভাইটাল সাইন`:`, `C/O`, `Dx`, `Rx`, `Adv`, `F/U` …) is mapped to one of 11 groups (COMPLAINT, HISTORY, VITAL, EXAM, INVEST, DIAG, TREAT, ADVICE, FOLLOW …) and combined with word shape and gazetteer marks. Output is post-processed so punctuation is always O and BIO sequences are always valid.

4.2 Evaluation protocol

Entity-level micro precision / recall / F1 with `seqeval` (exact span **and** type match). Two cross-validation schemes on our corpus: **random 5-fold by note**, and **domain-held-out 5-fold** (all notes of a clinical specialty held out together — e.g. the model never sees a dermatology note when predicting on dermatology), the closer proxy for the hidden set with unseen diseases and drugs.

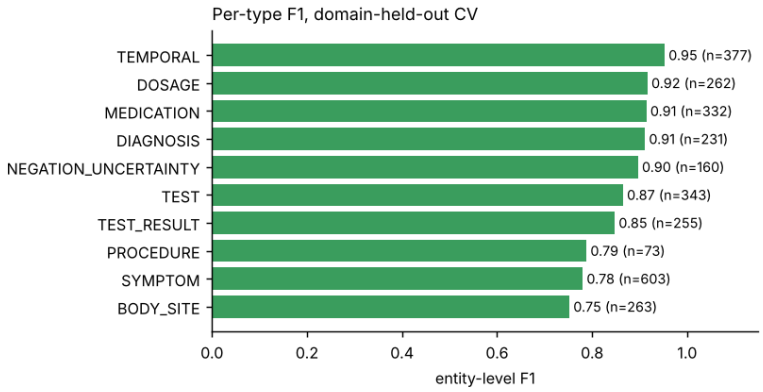
4.3 Results

evaluation	model	precision	recall	F1
random 5-fold (by note)	full: lexicon + section features	0.888	0.864	0.876
random 5-fold (by note)	no gazetteer lexicon	0.866	0.828	0.847
random 5-fold (by note)	no section features	0.893	0.862	0.877
random 5-fold (by note)	neither (plain CRF)	0.873	0.825	0.848
domain-held-out 5-fold	full: lexicon + section features	0.873	0.849	0.861
domain-held-out 5-fold	no gazetteer lexicon	0.836	0.792	0.814
domain-held-out 5-fold	no section features	0.873	0.841	0.857
domain-held-out 5-fold	neither (plain CRF)	0.849	0.780	0.813

- The gazetteer is the single most useful component (**4.7 F1 points** under domain hold-out): it lets the model recognise drug and disease names it never saw in training.
- Section context helps less (0.4 points) because many of our synthetic notes are short or unstructured; on the structured released notes we expect more.

Per-type F1, domain-held-out:

type	precision	recall	f1-score	support
DIAGNOSIS	0.909	0.913	0.911	231
SYMPTOM	0.783	0.778	0.780	603
MEDICATION	0.929	0.901	0.914	332
DOSAGE	0.910	0.924	0.917	262
TEST	0.879	0.851	0.865	343
TEST_RESULT	0.892	0.808	0.848	255
BODY_SITE	0.760	0.745	0.752	263
PROCEDURE	0.844	0.740	0.788	73
TEMPORAL	0.970	0.936	0.953	377
NEGATION_UNCERTAINTY	0.927	0.869	0.897	160
micro avg	0.873	0.849	0.861	2899
macro avg	0.880	0.847	0.863	2899



Per-type F1

Learning curve (fixed 40-note hold-out, growing training set) — the expansion mattered:

train_notes	train_tokens	precision	recall	F1
20	1065	0.843	0.775	0.808
40	2055	0.859	0.808	0.833
80	3973	0.891	0.847	0.868
120	6055	0.879	0.861	0.870
160	8306	0.914	0.894	0.904
192	10058	0.911	0.896	0.903

4.4 Error analysis

- **BODY_SITE and SYMPTOM** are the weakest (they are also the most convention-sensitive: where does the site end and the symptom begin?). Many errors are boundary errors, which `seqeval` counts as full misses.
- **TEST_RESULT** is weaker than TEST: qualitative results ("সামান্য বেশি", "প্রসারিত") are open-ended and unseen in training.
- Unseen vocabulary in *shorthand* complaints (`C/O:` রক্তে চর্বি বেশি ধরা পড়েছে) is often left untagged: the CRF is conservative on words it has not seen.

4.5 Predictions for the released notes

37619 entity spans on 1691 of 1691 released notes (22.2 per note; our corpus has 12.5 per note).

entity type	spans	distinct forms	notes containing	share %
SYMPTOM	9097	535	1686	24.2
TEMPORAL	4962	132	1691	13.2
BODY_SITE	4930	246	1618	13.1
MEDICATION	4178	252	1656	11.1
TEST	3933	324	1351	10.5
DOSAGE	3404	212	1612	9.0
TEST_RESULT	2988	798	1111	7.9
DIAGNOSIS	2455	286	1486	6.5
NEGATION_UNCERTAINTY	1432	8	999	3.8
PROCEDURE	240	25	213	0.6

Output formats (`outputs/`): `ner_predictions_bio.csv` (strict `note_id,token,tag`, validated by the official validator), `ner_predictions_spans.csv` (character offsets into `note_text`), `ner_predictions.jsonl`. The tokenizer is the same deterministic one used for training (whitespace split, edge punctuation split off, internal `- / .` kept, abbreviations such as মি.গ্রা. kept whole).

4.6 Why the hidden-test score will differ from our CV score

(i) Our data are synthetic and single-author; (ii) the hidden gold's conventions are unknown (units in `TEST_RESULT`? split of organ + symptom? history diagnoses?) — see "Known ambiguities" in `ANNOTATION_CONVENTIONS.md`; (iii) the released notes use richer vocabulary than ours. We expect a drop, mostly in BODY_SITE/SYMPTOM boundaries.

5. Stage 2 — Phenotype clustering

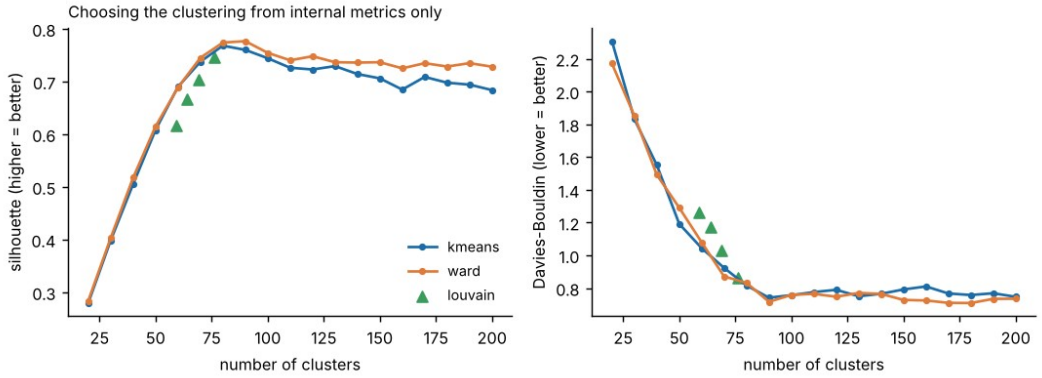
5.1 Representation (entities only)

For each of the 1691 released notes we use **only the entities predicted by the NER model** — not the note id, the file order, or any label:

- **View A:** DIAGNOSIS mentions → character 2–4-gram TF-IDF (robust to Bengali inflection and transliteration: আর্টিকারিয়া ≈ আমবাত (আর্টিকারিয়া)).
- **View B:** typed bag of DIAGNOSIS / SYMPTOM / MEDICATION / TEST / PROCEDURE / BODY_SITE phrases and their words → TF-IDF (entity co-occurrence). `DOSAGE`, `TEMPORAL`, `TEST_RESULT`, `NEGATION` are dropped (numbers and boiler-plate).
- Both views L2-normalised, weighted 1 : 1, concatenated, reduced to 64 dimensions (truncated SVD) and length-normalised.

5.2 Algorithm and model selection — internal metrics only

We compared k-means (k = 20…200), Ward agglomerative (k = 20…200) and Louvain communities on a cosine 10-NN graph (4 resolutions) and kept the configuration with the best **silhouette** (tie-break: lower **Davies–Bouldin**). No phenotype labels, no proxy labels were used for selection.



Internal metrics

Result: **ward with 90 clusters**, silhouette 0.777, DB 0.717; cluster sizes min 5, median 17, max 43. The silhouette curve rises steeply up to ≈ 80–90 clusters and then plateaus or declines, and Louvain independently finds 59–76 communities (depending on resolution), i.e. the notes contain on the order of 60–90 distinct condition groups.

5.3 Sanity check against the notes' own diagnosis line (not a score)

NMI = 0.905, ARI = 0.565 (on 1315 notes that print a diagnosis line)

This is **not** the hidden gold: the regex-extracted diagnosis strings are much finer-grained (203 distinct strings for 1315 notes) than a typical phenotype label, and our clusters use the same predicted DIAGNOSIS entities, so the check is partly circular. We report it only to show that the

unsupervised structure is not arbitrary. NMI is tolerant of over-clustering, ARI is not; if the hidden labels are much coarser than ~80 groups ARI will be lower than NMI.

5.4 What the clusters look like

cluster	size	top_diagnoses	top_medications
0	43	অ্যালার্জিক রাইনাইটিস (22); মৌসুমি অ্যালার্জিক রাইনাইটিস (10)	ফেক্সোফেনাডিন (15); ফ্লুটিকাসোন নাসাল স্প্রে (15)
1	37	আর্টিকারিয়া (20); তীব্র আর্টিকারিয়া (13)	ক্যালামাইন লোশন (27); সিটিরিজিন সিরাপ (15)
2	36	তীব্র ওটাইটিস মিডিয়া (18); ওটাইটিস মিডিয়া (9)	অ্যামোক্সিসিলিন সিরাপ (17); প্যারাসিটামল সিরাপ (17)
3	35	স্ক্যাবিস (35); খোসপাঁচড়া (14)	পারমেথ্রিন (14); সিটিরিজিন সিরাপ (13)
4	34	ভাইরাল সংক্রমণজনিত জ্বর (11); তীব্র ভাইরাল ফিভার (10)	প্যারাসিটামল (34); অ্যান্টিবায়োটিকের (25)
5	33	গোড়ালিব্যথা (10); প্লান্টার ফ্যাসাইটিস (10)	ডাইক্লোফেনাক জেল (33); প্যারাসিটামল (33)
6	31	অ্যাটোপিক ডার্মাটাইটিস (18); ডার্মাটাইটিস (6)	হাইড্রোকর্টিসোন ১% ক্রিম (19); ইমোলিয়েন্ট ক্রিম (15)
7	31	টনসিলাইটিস (17); শিশুর টনসিলাইটিস (8)	প্যারাসিটামল (19); অ্যামোক্সিসিলিন সিরাপ (12)

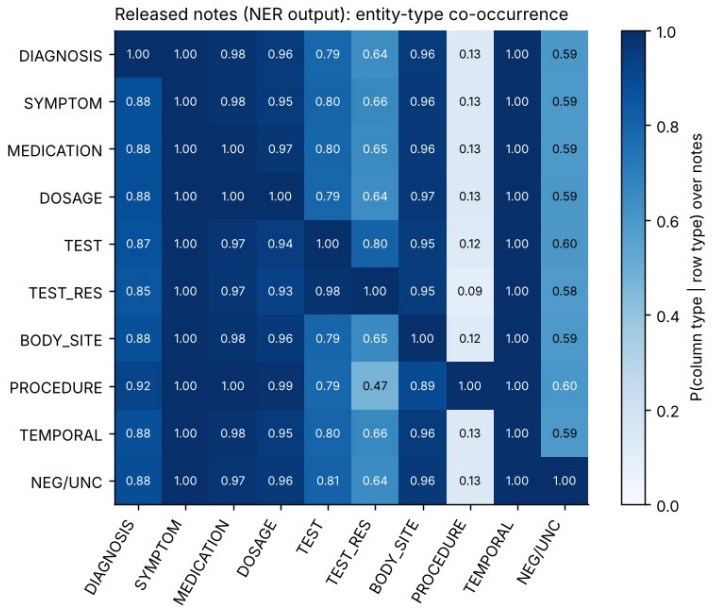
(Full table: `outputs/cluster_summary.csv`; assignment: `outputs/clusters.csv`.)

5.5 Frequency and relationship analysis on the NER output

- Frequency of predicted entities by type: `outputs/test_entity_frequency.csv`; most frequent entities per type: `test_top_entities.csv`.
- **Relationship analysis**: diagnosis → medication and diagnosis → symptom associations by within-note co-occurrence, lift and PMI (`test_diagnosis_medication_pairs.csv`, `test_diagnosis_symptom_pairs.csv`). Strongest diagnosis–drug pairs (highest lift among pairs co-occurring in ≥ 5 notes):

diagnosis	medication	co_notes	n_a	n_b	lift	pmi
তীব্র ল্যারিঞ্জাইটিস	কণ্ঠস্বরের সম্পূর্ণ	5	6	7	200.950	7.650
বসন্তকালীন অ্যালার্জিক চোখের প্রদাহ	ওলোপাটাডিন ০.১% আই ড্রপ	5	6	7	200.950	7.650
অ্যালার্জিক কনজাংটিভাইটিস	কিটোটিফেন ০.০৫% আই ড্রপ	5	9	5	187.560	7.550
এন্টেরোবিয়াসিস	আয়রন ও ফলিক অ্যাসিড সিরাপ	5	5	9	187.560	7.550
কৃমি	আয়রন ও ফলিক অ্যাসিড সিরাপ	5	5	9	187.560	7.550
শিশুর ত্বকের ব্যাকটেরিয়াল সংক্রমণ	মুপিরোসিন ২% মলম	5	5	9	187.5	7.55

diagnosis	medication	co_notes	n_a	n_b	lift	pmi
					60	0
মেলাজমা	ভিটামিন সি সিরাম	8	8	10	168.800	7.400
তীব্র অ্যালার্জিক প্রতিক্রিয়া	ফেনিরামিন ইনজেকশন	6	6	10	168.800	7.400
হাঁটুর বাত	গ্লুকোসামিন ও কনড্রয়েটিন সালফেট	6	9	7	160.760	7.330
মেলাজমা	ট্রেটিনোইন ০.০২৫% ক্রিম	8	8	11	153.450	7.260



Entity-type co-occurrence on the released notes

6. Stage 3 — Retrieval-augmented question answering

6.1 Knowledge base

Every note becomes a document with (a) normalised word keys, (b) the set of clinical concepts it mentions, (c) age and sex parsed from the header, (d) its entity spans — gold spans for our notes, **predicted** spans for the released notes.

6.2 Retrieval

1. **Normalisation.** Spelling skeleton (drops hasanta, collapses ি/ী ু/ূ ণ/ন শ/ষ/স, অ্যা→এ) + light Bengali stemming, so মেটফর্মিন = মেটফরমিন and ডায়াবেটিসের = ডায়াবেটিস.
2. **Concept table** (`concepts.py`, 73 concepts / 411 surface variants): bilingual synonyms, abbreviations (T2DM, NS1, PPI, SSRI, UTI, BPH …) and drug classes (SSRI → এসসিটালোপ্রাম,

সার্ট্রালিন …). Multi-word concepts also match **order-free inside a window** (রাতে বেশি চুলকানি ≈ রাতে চুলকানি).

3. **Polarity-aware matching.** Concepts are detected per *clause*; a concept in a negated clause (ডায়াবেটিস নেই) or about a family member (মায়ের ডায়াবেটিস) does **not** count for the patient — unless the query itself asks for the negation (অ্যান্টিবায়োটিকের প্রয়োজন নেই).
4. **Constraints.** Child / elderly / sex from the note header, numeric thresholds (`HbA1c ৯-`এর বেশি is evaluated on the value in the note).
5. **Scoring.** concept coverage (0.55) + IDF-weighted coverage of remaining rare query terms (0.25) + BM25 (0.20, tie-breaker); constraints multiply the score by 0.3 when violated; a +0.10 bonus when the queried disease sits in a DIAGNOSIS span inside the note's diagnosis section (নির্ণয়: / `Dx:`) — the note is *about* it, not merely mentioning it in the history. All conditions satisfied → exact set; otherwise the closest partial matches are shown and **labelled** with the missing condition.

6.3 Answer generation — grounded by construction

No generative model is used. The answer is assembled from templates, entity strings and verbatim sentence fragments copied from the retrieved notes, each cited with `[note-id]`: a count and, depending on the query intent, an aggregated list (drugs / symptoms / findings / treatment with the number of notes), then per-patient lines (age, sex, diagnosis, drugs with dosage, evidence excerpt). If nothing satisfies the query, the system says so and refuses to guess.

6.4 Results on the 30 released queries

id	query	detected concepts	all conditions met	partial shown	team-only KB
q01	ডায়াবেটিস রোগীদের জন্য কোন কোন ওষুধ নথিভুক্ত …	diabetes	89	0	17
q02	নিউমোনিয়ায় আক্রান্ত রোগীদের ক্ষেত্রে কোন কোন …	pneumonia	37	0	3
q03	হার্ট ফেইলিওর আছে এমন রোগীদের বর্ণনা দিন। …	heart_failure	4	0	4
q04	হাঁপানি আছে ও রাতে কাশি হয় এমন রোগীদের জন্য কো…	asthma, cough_night	34	0	5
q05	গাউট এবং সিরাম ইউরিক অ্যাসিড বেশি আছে এমন রোগী…	gout, uric_acid	26	0	1
q06	উচ্চ রক্তচাপ ও ডায়াবেটিস আছে এমন রোগীদের মধ্য…	diabetes, hypertension	3	0	3
q07	কোন কোন রোগীর শ্বাসকষ্ট আছে?…	dyspnea	190	0	18
q08	পাতলা পায়খানা ও জল শূন্যতা আছে, ওআরএস নিচ্ছে …	dehydration, diarrhea, ors	23	0	1
q0	ডায়াবেটিক রোগীদের বিস্তারিত তথ্য এনে	diabetes	89	0	17

id	query	detected concepts	all conditions met	partial shown	team-only KB
9	দিন। …				
q10	হার্ট ফেইলিওর, ডায়াবেটিস ও কিডনির সমস্যায় আক্…	diabetes, heart_failure, loop_diuretic	1	3	1
q11	পেপটিক আলসার ও পেটের ওপরের অংশে জ্বালাপোড়া আছ…	epigastric_burning, nsaid, peptic_ulcer, ppi	2	8	2
q12	অ্যালার্জিক রাইনাইটিস ও হাঁচি আছে এবং অ্যান্টি…	allergic_rhinitis, antihistamine, sneezing	36	0	1
q13	কোন কোন রোগীর ডেঙ্গু ও জ্বর আছে এবং NS1 পরীক্ষ…	dengue, fever, ns1, platelet	30	0	4
q14	স্ক্যাবিস (খোসপাঁচড়া) ও রাতে চুলকানির সমস্যা র…	night_itch, permethrin, scabies	14	0	1
q15	বিষণ্নতা আছে, ক্লান্তি বা দুর্বলতার কথা জানিয়…	depression, fatigue, ssri	7	0	1
q16	সেপসিস, জ্বর ও বিভ্রান্তি (কনফিউশন) আছে এবং রক…	blood_test, confusion, ct_scan, fever, sepsis	1	0	1
q17	হাইপারকোলেস্টেরোলেমিয়া আছে, অ্যাটরভাস্টাটিন ন…	hypercholesterolemia, lifestyle, statin	6	0	0
q18	বুকে ব্যথা আছে, ইসিজি বা ইকো (ইকোকার্ডিওগ্রাম)…	chest_pain, ecg, echo, ischemic_heart	3	0	3
q19	মেটফর্মিন চলছে এবং HbA1c ৯-এর বেশি এমন T2DM রো…	diabetes, hba1c, metformin	8	0	2
q20	নিউমোনিয়া ও শ্বাসতন্ত্রের সংক্রমণ আছে, অ্যামো…	amoxicillin, antibiotic, cxr, dyspnea, pneumonia, resp_infection	0	8	0
q21	তীব্র আর্টিকারিয়া আছে এমন রোগীদের জন্য কোন কো…	urticaria	45	0	2
q22	ভাইরাল জ্বরের কোন কোন রোগীকে বলা হয়েছে যে অ্য…	antibiotic, viral_fever	54	0	4
q23	প্রস্টেট বৃদ্ধি (BPH) আছে এবং ট্যামসুলোসিন নিচ…	bph, tamsulosin	15	0	1
q24	দীর্ঘস্থায়ী কাশি, ওজন কমা ও রাতে ঘাম দেয় এবং …	chronic_cough, night_sweat, tb, weight_loss	5	0	1
q25	জ্বর ও সর্দি-কাশি নিয়ে আসা শিশুদের মধ্যে কাদে…	fever, paracetamol_syrup	33	0	1
q26	হারপিস জোস্টার ও যন্ত্রণাদায়ক ফোসকা আছে এবং অ…	acyclovir, blister, herpes_zoster	23	0	2

id	query	detected concepts	all conditions met	partial shown	team-only KB
q27	মাইগ্রেন আছে, বমি বমি ভাব বা বমি এবং আলোতে অস্…	migraine, nausea_vomiting, photophobia	2	8	2
q28	টাইফয়েড ও দীর্ঘদিনের জ্বর আছে এমন রোগীদের খুঁ…	antibiotic, prolonged_fever, typhoid	4	0	1
q29	PCOS আছে, মাসিক অনিয়মিত এবং মেটফর্মিন নিচ্ছেন…	irregular_periods, metformin, pcos	17	0	2
q30	প্রস্রাবে জ্বালাপোড়া আছে, urine R/E করা হয়েছ…	antibiotic, dysuria, urine_re, uti	20	0	2

All 30 answers: `outputs/rag_answers.md` (readable), `.csv`, `.jsonl`. Representative example (q19, numeric constraint):

নথিতে ৮ জন রোগীর নোট পাওয়া গেছে যেখানে প্রশ্নের সব শর্ত মিলেছে।
রোগীভিত্তিক বিবরণ (উৎস নোট সহ):
- [n048] ৫০ বছর, পুরুষ — রোগ নির্ণয়: টাইপ ২ ডায়াবেটিস মেলিটাস — ওষুধ: মেটফর্মিন ৫০০ মি.গ্রা. দিনে ২ বার খাবারের পর, গ্লিমেপিরাইড ১ মি.গ্রা. সকালে ১টি — নোট থেকে: «খালি পেটে রক্তে শর্করা ১৮৫ মি.গ্রা./ডেসিলিটার, HbA1c ৯.২% | HbA1c ৩ মাস পর পুনরায় পরীক্ষা করুন»
- [n192] ৫২ বছর, পুরুষ — রোগ নির্ণয়: ডায়াবেটিস, অনিয়ন্ত্রিত টাইপ ২ ডায়াবেটিস — ওষুধ: মেটফর্মিন ১ গ্রাম দিনে ২ বার, ইনসুলিন রাতে ১০ ইউনিট — নোট থেকে: «HbA1c ১০.২%, খালি পেটে শর্করা ২১০ মি.গ্রা./ডেসিলিটার | HbA1c ৩ মাস পর»

Generalisation test (queries not in the released set) — the concept table was written with the public queries in view, so we checked unseen questions; unknown concepts fall back to BM25 + fuzzy stems:

unseen query	concepts found	all-condition matches	partial	top notes
হাইপোথাইরয়েডিজম আছে এবং লিভোথাইরক্সিন নিচ্ছেন এমন রোগীদের দেখান।	(none → BM25 fallback)	2	0	n209 n050
শিশুদের কানের সংক্রমণে কোন কোন ওষুধ দেওয়া হয়েছে?	(none → BM25 fallback)	8	0	MB-0326 MB-0351 MB-0602 MB-0094
অ্যানিমিয়া বা রক্তস্বল্পতা আছে এমন গর্ভবতী মহিলাদের তথ্য দিন।	(none → BM25 fallback)	10	0	n204 n197 n059 MB-0831
কোন রোগীদের কোভিড-১৯ হয়েছে?	(none → BM25 fallback)	1	0	n079

unseen query	concepts found	all-condition matches	partial	top notes
ইনফ্লুয়েঞ্জা বা ভাইরাল জ্বরে কোন কোন ওষুধ দেওয়া হয়েছে?	viral_fever	102	0	MB-1064 MB-0612 n116 MB-0118

6.5 The "own annotated corpus only" question

The rule says retrieval must draw only from the team's own annotated corpus. Our 232 notes alone can answer only part of the 30 queries (a handful of notes per condition). The queries, however, ask which *patients* have a condition, which presupposes the released patient notes. We therefore treat "annotated corpus" as *our notes + the released notes annotated by our own NER model* in the default run, and we additionally ship the strict variant (`rag_answers_team_corpus_only.*`, 28 of the 30 queries return at least one note). **We will open an issue / ask the organisers which interpretation is intended**; switching is one argument (`ask --team-only`, or `release_df=None`).

6.6 Limitations of the RAG stage

- Extractive, template-based Bengali — useful and verifiable, but not fluent prose. A local LLM could polish wording but cannot be shipped in an offline, exactly-reproducible repo; if allowed, it should be constrained to rephrase only the retrieved text.
- Answers inherit NER errors for the released notes (e.g. a negated drug mentioned in advice can appear as "recorded"). Evidence excerpts are verbatim, so the reader can always verify.
- The concept table is hand-written; new abbreviations need new entries (fallback: BM25 + stems).
- No clinician evaluation yet; "correctness/usefulness" in the clinical advisor's rubric is untested by us.

7. Reproducibility

`python scripts/run_pipeline.py` (or *Restart & Run All* in the notebook) regenerates every file in `outputs/`. All randomness is seeded (CRF is deterministic; k-means, SVD, Louvain take `random_state=42`). Library versions are pinned in `requirements-lock.txt` and recorded with MD5 hashes of all outputs in `outputs/run_manifest.json`. `scripts/check_reproducibility.py` re-runs the pipeline with a different `PYTHONHASHSEED` and asserts that all hashes are unchanged. **Verified (2026-11-11):** the notebook run and a second, independent full run of `scripts/run_pipeline.py` under `PYTHONHASHSEED=12345` produced byte-identical files for all 27 output files in `outputs/` (`scripts/check_reproducibility.py`: "REPRODUCIBLE"). The check also caught and led us to fix one hash-order dependence (tie ordering in the association tables).

8. Limitations, risks and next steps

1. **Annotation quality** is the dominant risk: synthetic, single-author, no clinician review, no IAA. *Next step:* clinician spot-check of ≈ 50 notes; resolve the ambiguities listed in the conventions file.
2. **Domain shift** between our notes and the released notes (vocabulary, length). *Next step:* add notes in the observed registers (needs more synthetic authoring, not the released text).
3. **NER model class.** A CRF cannot use cross-sentence context or pretrained knowledge. *Next step (if offline weights are allowed):* IndicBERT/MuRIL token classifier initialised from public weights, ensembled with the CRF.
4. **Clustering granularity** is inferred from internal metrics; hidden label granularity is unknown.
5. **RAG** is extractive and concept-table dependent.

9. Questions for the organisers

1. Which interpretation of "own annotated corpus" is intended for RAG (§6.5)?
2. What exact format is expected for NER predictions (we provide BIO per token, spans with character offsets, and JSONL)?
3. Do TEST_RESULT spans include units; are organ + symptom phrases split?

Appendix A — Repository map

Path	Content
`MedBhasha_Pipeline.ipynb`	the single runnable notebook (executed, outputs embedded)
`src/medbhasha/`	library: `tokenizer`, `corpus`, `lexicon`, `ner`, `evaluation`, `cluster`, `concepts`, `rag`, `analysis`, `run_all`, `ask`
`data/round1/`	annotated corpus (BIO), raw text, metadata, annotation sources (span markup), the 40 submitted Round-1 notes
`data/round2/`	organiser-released test notes and queries (unchanged, with BOM removed)
`outputs/`	all generated results + `run_manifest.json`
`round1_deliverables/`	validation report, source declaration, EDA summary, Round-1 fix log
`docs/`	this report, `ANNOTATION_CONVENTIONS`
`resources/`	organiser files used unchanged (`validate.py`, schema, rules, worked example)

Path	Content
`scripts/`	`build_corpus.py`, `run_pipeline.py`, `check_reproducibility.py`, `make_notebook.py`, `build_docs.py`, `make_zip.py`, `serve_app.py`
`tests/`	sanity tests